

CS-1153
Examination – Nov/Dec 2016
B.E. V Sem: Computer Science Engg.
Data Base Management System

Time: 3 Hrs

Max. Marks: 70

Min. Marks: 22

Note: Total number of questions 10. Attempt one question from each unit (including all parts). Assume missing data, if any, suitably.

- Q1.(A)** Define DBMS. Unit-I
- (B) What are responsibilities of a DBA? 2
- (C) Describe three level of data abstraction. 2
- (D) What is an ER diagram? Construct an ER diagram for a hospital with a set of patients and set of doctors. associate with each patients and a set of doctors associate with each patient a log of the various tests and examination conducted 3

OR

- Q2.(A)** Discuss in detail about database system architecture with neat diagram. 7
- (B) Explain the following with their advantages and disadvantages.
 a. Hierarchical database model b. Network database model c. Relational database model 7

Unit-II

- Q3.(A)** Discuss the basic operations that can perform using relational algebra. Also explain the concept of referential integrity in database management system. 7
- (B) What is foreign key constraint? What are various options associated with is like cascaded deletion etc. Take suitable example to demonstrate. 7

OR

- Q4.(A)** Explain the distinctions among the terms primary key, candidate key and super key. 4
- (B) Consider the following employee database, where the primary keys are underlined.
 Employee (employee-name, street, city)
 Works (employee-name, company-name, salary)
 Company (company-name, city)
 Managers (employee-name, manager-name)
 Give an expression in SQL for each of the following queries. 10

- (i) Find the names of all employees who work for First Bank Corporation and live in Las Vegas.
- (ii) Find the names, street addresses and cities of residences of all employees who work for First Bank Corporation and earn more than \$10000.
- (iii) Find all employees who do not work for First Bank Corporation.
- (iv) Find the company that has the smallest payroll.
- (v) Find all employees in the database who do not live in the same cities and on the same streets as do their managers.

UNIT-III

- Q5.(A)** Why the normalisation process is necessary for a good database design? Discuss in detail the Boyce-Codd normal form with suitable example. 7
- (B) Explain the Lossless join decomposition with a suitable example. 7

OR

- Q6.(A) Explain why 4NF is a normal form more desirable than BCNF. 7
- (B) Consider the relation schema R (A,B,C), which has the FD B $\rightarrow$ C. If A is a candidate key for R, is it possible for R to be in BCNF? If so, under what conditions? If not, explain why? 7

Unit-IV

- Q7.(A) What are the basic properties of a transaction? Explain these properties with the help of examples. 7
- (B) Discuss the different measures of transaction equivalence. What is the difference between conflict equivalence and view equivalence? 7

OR

- Q8.(A) Database-system implementers have paid much more attention to the ACID properties than have file-system implementers. Why might this be the case? 7
- (B) What is a schedule (history)? Explain the distinction between the terms serial schedule and serializable schedule. 7

Unit-V

- Q9.(A) What are the different advantages of a distributed database management system? Discuss in detail the different functions of a DDBMS to manage the complexity of data distribution. 7
- (B) Explain the difference between OODBMS and DBMS. 7

OR

- Q10.(A) What is the use of data warehouse? Discuss the architecture of data warehouse and its functionality in detail. 7
- (B) write short notes on any three of the following : 7
- (1) Web database
 - (2) Temporal database
 - (4) Data mining techniques
 - (5) mobile databases

AI-2042
Examination –May-June - 2022
B.Tech. IV Sem : AIADS
Database Management System

Time : 3 Hrs

Max. Marks : 70

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- Q.1 (a) Discuss Data Independence 02
- (b) Define Entity and Relationship in ER model. 02
- (c) Define instance and schema of database. 03
- (d) Explain Enhanced ER diagrams with specialization and aggregation constraints. 07

OR

Analyze and find the ER model for the School System? 07

- Q.2 (a) Define aggregate function in SQL. 02
- (b) Write syntax for select commands in SQL. 02
- (c) Define join operator in Relational algebra. 03
- (d) Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. Calculate the minimum number of tables required to represent this situation in the relational model? 07

OR

Consider given relations

employee(name,salary,deptno)

department (deptno, deptname, address)

Write the following queries in relational algebra/SQL.

- Find the names of employees who have worked for the 'CSE' department.
- Find the average salary of employees working in the 'CSE' department.
- Find the department name paying highest salary to their employee.
- List the department details.

- Q.3 (a) Define functional dependency. 02
 (b) Differentiate super key and candidate key 02
 (c) What is the minimal set of Functional dependency? 03
 (d) Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A, AC \rightarrow B\}$. Write the candidate keys of R ? 07

OR

Consider the following relationship: $R (W, X, Y, Z)$ and the following dependencies: 07
 determine the present normal form and also normalize in further higher normal form.

$W \rightarrow XYZ$

$XY \rightarrow WZ$

$Z \rightarrow X$

- Q.4 (a) Define Transaction. 02
 (b) What are the states of transaction? 02
 (c) Explain ACID properties. 03
 (d) Consider the transactions $T1, T2$, and $T3$ and the schedules $S1$ given below. $T1$: 07
 $r1(X); r1(Z); w1(X); w1(Z)$
 $T2: r2(Y); r2(Z); w2(Z)$
 $T3: r3(Y); r3(X); w3(Y)$ Check schedule $S1$ for serializability.

OR

Discuss concurrency protocols. Which of the following concurrency control protocols 07
 ensure both conflict serializability and freedom from deadlock.

- Q.5 (a) Discuss the Primary and Secondary indexes. 02
 (b) Define Tree Indexing. 02
 (c) Compare and Contrast Extendible Hashing with Linear Hashing. 03
 (d) Explain B+ trees? Discuss about this Dynamic Index Structure. 07

OR

Illustrate insertion and deletion of an element in B+ trees with an example. 07

Data Base Management System

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UNIT-III

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Unit-IV

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AI-2042

Examination – April/May- 2023

B.Tech. IV Sem : Artificial Intelligence & Data Science
Data Base Management System

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

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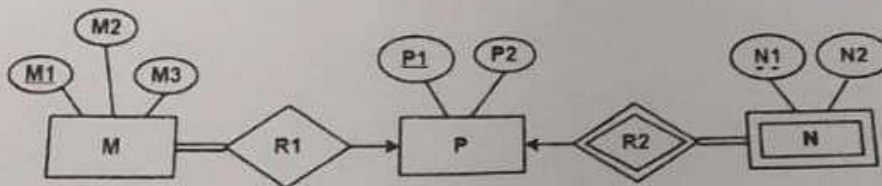
Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) Define DBMS? List Database Applications. 02
- (b) Explain Database Administrator's responsibilities and the main functions of Database Administrator. 02
- (c) List five significant differences between a File processing system and a DBMS. 03
- (d) Explain Data Abstraction in DBMS? Explain different levels of data abstraction in a DBMS. 07

OR

Define Database Architecture with appropriate diagram. 07

- Q.2 (a) Find the minimum number of tables required for the following ER diagram in relational model- 02



02

03

07

- (b) Describe Relational Model and its concepts in DBMS.
- (c) Explain SQL and the different languages present in SQL.
- (d) Explain relational algebra and types of relational Operation.

OR

07

Write all the rules for Relational database design using ER to relational mapping.

02

Explain normalization of relations.

02

Write a short note on Dependency Preserving.

AI-2042

- (c) Explain trivial and Non-trivial functional dependency.
- (d) Discuss in details about Relational decomposition and types of decomposition.

OR

Describe Normal forms and its type with example.

- Q.4
- (a) What is concurrency control?
 - (b) Define database recovery concepts.
 - (c) What is a lock? Explain the major difference between a shared lock and an exclusive lock during a transaction in a database.
 - (d) Explain different states of transactions with Diagram.

OR

Briefly discuss on the two phase locking protocol used in concurrency control.

- Q.5
- (a) Explain indexing.
 - (b) Write Query Evaluation Plan.
 - (c) Explain data Storage and types of storage devices in data storage.
 - (d) Describe B Tree with an example and properties of B Tree.

OR

Define Query Processing with Block Diagram of Query Processing.

CS-1843

Examination – May-June.- 2022
B.Tech. IV Sem: Computer Science & Engineering
Data Base Management System

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

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- Q.1 (a) Write down the advantages and disadvantages of DBMS. 02
- (b) Discuss three level schema architecture of DBMS. 02
- (c) Differentiate between strong entity set and weak entity set? Draw an ER diagram to illustrate weak entity set? 03
- (d) Develop an ER diagram for keeping track of information about a Hospital database taking into accounts at least five entities. 07

OR

Compare generalization, specialization and aggregation with suitable example. 0

- Q.2 (a) What do you understand by Union Compatibility? 0
- (b) Differentiate between procedural and non procedural language. 0
- (c) Write down the DDL comments in SQL. 0
- (d) Consider the following database schema
 Patient(P_id, P_name, P_gender, P_age, P_disease)
 Doctor(D_id, D_name, D_specialist)
 Treatment(P_id, D_id, Date)
 Solve the following queries in Relational Algebra
- Get the name of the female patients who is suffering from heart disease and having age below 50
 - Get the name of the doctor who gave the treatment for the cancer disease.
 - Get the name of the patient who got treatment on 5th May 2022.
 - Get the name of the doctor who has not given treatment on any patient.
 - Get the name of the patients who got treated by the doctor who cardiologist.

OR

Consider the following database schema

EMP(Name,SSN,Salary,SuperSSN,Gender,Dno) DEPT(DNum,Dname,MgrSSN,Dno)
DEPT_LOC(Dnum,Dlocation)
DEPENDENT(ESSN,Dep_name)
WORKS_ON(ESSN,Pno,Hours) PROJECT(Pname,Pnumber,Plocation,Dnum)

Write the SQL queries for the following

- Retrieve the name, address, salary of employees who work for the Research department.
- Find the names of employees who work on all projects controlled by department number 4.
- Retrieve the SSN of all employees who either in department no :4 or directly supervise an employee who work in department number :
- Retrieve the names of employees who have no dependents
- Retrieve each department number, the number of employees in the department and their average salary.

- Q.3 (a) What are the various types of anomalies associated with the database? (2) 02
- (b) Given below are two sets of FD's for a relation R(A,B,C,D,E).Are they equivalent? 02
F={A→C,AC→D, E→AD,E→H} and G={A→CD,E→AH} (2)
- (c) Consider a schema R (A, B, C, D) and functional dependencies A → B and C →D. Find 03
whether the decomposition of R into R1 (A, B) and R2(C, D) belongs to which one or both (dependency preserving and loss less join)? (3)
- (d) Consider the relation schema R(A,B,C,D,E,F) and the functional dependencies A→B,C→DF,AC→E, D→F. What is the primary key of this relation R? What is its highest 07
normal form? Preserving the dependency, decompose R into third normal form (7)

OR

What do you understand by multivalued dependency? Explain the 4NF with example. 07

- Q.4 (a) When does two operations in schedule are said to be conflict? (2) 02
- (b) Differentiate between Two phase locking and Rigorous two-phase locking. (2) 02
- (c) What are the ACID properties of a transaction? (3) 03
- (d) Briefly discuss on the two phase locking protocol used in concurrency control. How 07
does it guarantee serializability? (5)

OR

Explain the deferred and immediate modification versions of the log based recovery 07
scheme. Also Explain the purpose of checkpoint mechanism.

- Q.5 (a) Differentiate between the single and multi level indexing. (1) 02
- (b) How the B trees differ from the B+ tree. (2) 02
- (c) Explain the structure of B+ tree. (3) 03
- (d) What is the objective of query processing? What are the main phases of query 07
processing, explain.

OR

Construct a B+-tree in which the maximum number of keys in a node is 4 with the 07
following data. {12, 13, 17, 19, 23, 30, 45, 56, 71} Illustrate the stepwise insertion process. (5)

ICB-2143
Examination –April/May- 2023
B.Tech. IV Sem : CSE-ICB
Database Management System

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

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- Q.1 (a) List the database Applications? 02
 (b) Discuss Data Independence? 02
 (c) List out any three salient features of a database system. 03
 (d) Discuss about ER and EER with suitable examples. 07

OR

Discuss the Constraints in RDBMS. 07

- Q.2 (a) Define instance and schema? 02
 (b) For the SQL query, SELECT A, B FROM R WHERE B='apple' AND C = 'orange' on the table R(A, B, C, D), where A is a key, write any two equivalent relational algebra expressions. 02
 (c) When is a multivalued composite attribute used in ER modeling? 03
 (d) Write SQL DDL statements for the following (Assume suitable domain types): 07
 i. Create the tables STUDENT(ROLLNO, NAME, CLASS, SEM, ADVISER),
 FACULTY(FID, NAME, SALARY, DEPT). Assume that ADVISER is a foreign key
 referring to the FACULTY table.
 ii. Delete department with name 'CS' and all employees of the department.
 iii. Increment salary of every faculty by 10%.

OR

employee(employee-name, street, city) 07

works(employee-name, company-name, salary)

company(company-name, city)

manages(employee-name, manager-name)

(i) Find the names, street addresses, and cities of residence for all employees who work for the Company 'RIL Inc.' and earn more than \$10,000.

(ii) Find the names of all employees who live in the same cities as the companies for which they work.

(iii) Find the names of all employees who do not work for 'KYS Inc.'. Assume that all people work for exactly one company.

(iv) Find the names of all employees who earn more than every employee of 'SB Corporation'. Assume that all people work for at most one company.

- Q.3 (a) Discuss types of functional dependency. 02
- (b) What are the advantages of normalization? 02
- (c) Compare and contrast BCNF with 3NF. 03
- (d) Consider a schema R (A, B, C, D) and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Solve and find whether the decomposition of R into R1 (A, B) and R2(C, D) belongs to which one or both (dependency preserving and lossless join)? 07

OR

Consider the un-normalized relation R(A, B, C, D, E, F, G) with the FDs $A \rightarrow B$, $AC \rightarrow G$, $AD \rightarrow EF$, $EF \rightarrow G$, $CDE \rightarrow AB$. Trace the normalization process to reach 3NF relations. 07

- Q.4 (a) What is the transaction life cycle? 02
- (b) Discuss ACID property. 02
- (c) Outline the concept of theta-join. 03
- (d) What is meant by the lost update problem? 07

OR

Consider the transactions T1, T2, and T3 and the schedules S1 and S2 given below. 07

T1: $r1(X); r1(Z); w1(X); w1(Z)$, T2: $r2(Y); r2(Z); w2(Z)$, T3: $r3(Y); r3(X); w3(Y)$

S1: $r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)$

S2: $r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)$

Analyze which one of the schedules is conflict-serializable?

- Q.5 (a) What is a dense index? 02
- (b) Discuss about data on External storage. 02
- (c) Compare B and B+ tree. 03
- (d) Consider an EMPLOYEE file with 10000 records where each record is of size 80 bytes. The file is sorted on employee number (15 bytes long), which is the primary key. Assuming unspanned organization and block size of 512 bytes compute the number of block accesses needed for selecting records based on employee number if, 07
- No index is used
 - Single level primary index is used
 - Multi-level primary index is used
- Assume a block pointer size of 6 bytes.

OR

A clustering index is defined on the fields of which type? Analyze them 07
